



TEST CODE **02107032**

FORM TP 2014142

MAY/JUNE 2014

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

C A R I B B E A N A D V A N C E D P R O F I C I E N C Y E X A M I N A T I O N ®

BIOLOGY

UNIT 1 – Paper 032

ALTERNATIVE TO SCHOOL-BASED ASSESSMENT

2 hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. You may use a silent non-programmable calculator.
4. You are advised to take some time to read through this paper carefully.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Please begin Question 1 FIRST.

1. (a) (i) You are required to carry out a semi-quantitative test for glucose on an unknown solution, Z.

You are provided with the following five solutions in test tubes:

Test Tube A containing 2 cm³ of distilled water

Test Tube B containing 2 cm³ of 0.1% glucose

Test Tube C containing 2 cm³ of 0.5% glucose

Test Tube D containing 2 cm³ of 1.0% glucose

Test Tube E containing 2 cm³ of the unknown solution, Z

PROCEDURE

1. Add 2 cm³ of Benedict's solution to EACH of the test tubes labelled A, B, C, D and E.
2. Carefully shake each test tube and note the initial colour of the mixture. Record your observations in the appropriate column in Table 1.
3. Place the five test tubes in a heated water-bath to bring gently to a boil.
4. Remove the test tubes from the water-bath.
5. Note the final colour and record your observations in the appropriate column in Table 1.

TABLE 1: RESULTS OF EXPERIMENT

Test Tube	Solution	Initial Colour	Final Colour
A	Distilled water		
B	0.1% Glucose		
C	0.5% Glucose		
D	1.0% Glucose		
E	Solution Z		

[5 marks]

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- (ii) Based on your results, deduce the glucose concentration in Solution Z. Justify your answer.

Concentration of Solution Z: _____

Justification: _____

[3 marks]

- (b) Starting with a 10% glucose stock solution, calculate the volume of stock solution (cm^3) and distilled water (cm^3) that must be combined to give 100 cm^3 of 0.5% glucose solution. **Show your working.**

Volume of stock: _____

Volume of water: _____

[2 marks]

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- (c) Figure 1 shows the effect of temperature on the activity of a particular enzyme.

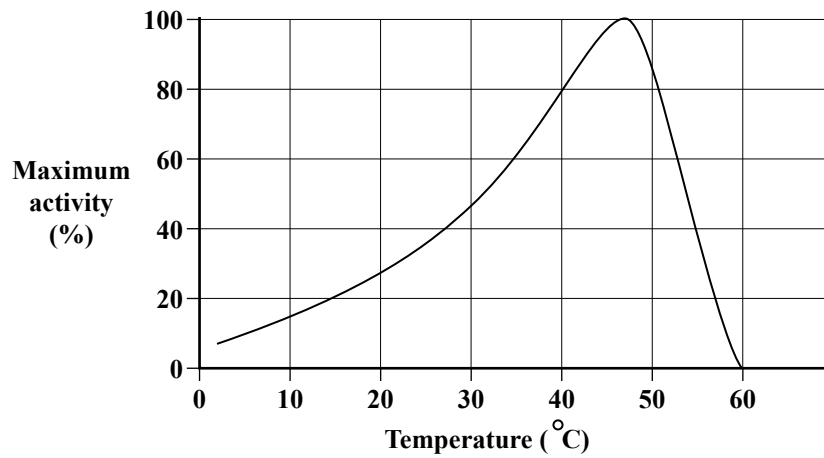


Figure 1. Effect of temperature on the activity of an enzyme

- (i) Describe the effect of temperature on the activity of this enzyme.

[3 marks]

- (ii) Determine the optimum temperature for the activity of the enzyme, based on the information provided in Figure 1.

[1 mark]

- (iii) Give an explanation for the steep decline in enzyme activity as temperatures become very high.

[2 marks]

Total 16 marks

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2. (a) Figure 2 is a photomicrograph of onion root tip cells, showing four key stages of mitosis.



I

II

III

IV

Figure 2. Photomicrograph of onion root tip cells showing stages of mitosis

<http://www.marietta.edu/~biol/introlab/Onion%20root%20mitosis.pdf>

- (i) Identify the stages labelled I, II, III and IV and for EACH stage, state ONE observable feature which justifies your identification.

Identity of Stage **I**: _____

Justification: _____

Identity of Stage **II**: _____

Justification: _____

Identity of Stage **III**: _____

Justification: _____

Identity of Stage **IV**: _____

Justification: _____

[8 marks]

- (ii) Name TWO features that would be clearly observed during mitosis if Stage IV is examined at a higher magnification of a compound microscope.

[2 marks]

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- (b) Seed colour in a plant species is determined by a single gene, Y, such that YY = yellow, Yy = yellow and yy = green. The expected and observed numbers of seeded offspring are given in Table 2.

TABLE 2: EXPECTED AND OBSERVED NUMBERS OF SEEDED OFFSPRING

Colour	Expected Number	Observed Number
Yellow	375	380
Green	120	120

- (i) Calculate the Chi-square value using the following formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Show your working.

Value: _____ **[3 marks]**

- (ii) If the Chi-square value for one degree of freedom at $p = 0.05$ is 3.84, what conclusions can be made, based on the results obtained in (b) (i) above.

_____ **[3 marks]**

Total 16 marks

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3. (a) Specimen A is a section of an entire ovary of a mammal. Examine the section under a low power objective of the compound microscope. In the space below, make a labelled, plan drawing of the specimen to show accurately the shape of the ovary and the distribution of tissues in it. State the magnification of your drawing.

[7 marks]

- (b) Table 3 shows data for changes in the lipid and sugar content of seeds during germination in the dark.

TABLE 3: LIPID AND SUGAR CONTENT OF SEEDS

Time From Sowing (days)	Mass per 100 Seeds (g)	
	Lipid	Sugar
2	26	2
4	24	6
5	16	11
6	12	20
7	8	23
8	5	24
10	3	20
11	2	18

- (i) On the grid provided on page 9, draw line graphs to illustrate the data in Table 3.
[5 marks]

- (ii) Describe the trend for changes in **sugar** content of the seeds shown in the graph.

[2 marks]

- (iii) Suggest reasons for the changes in lipid and sugar content of the seeds as germination progresses.

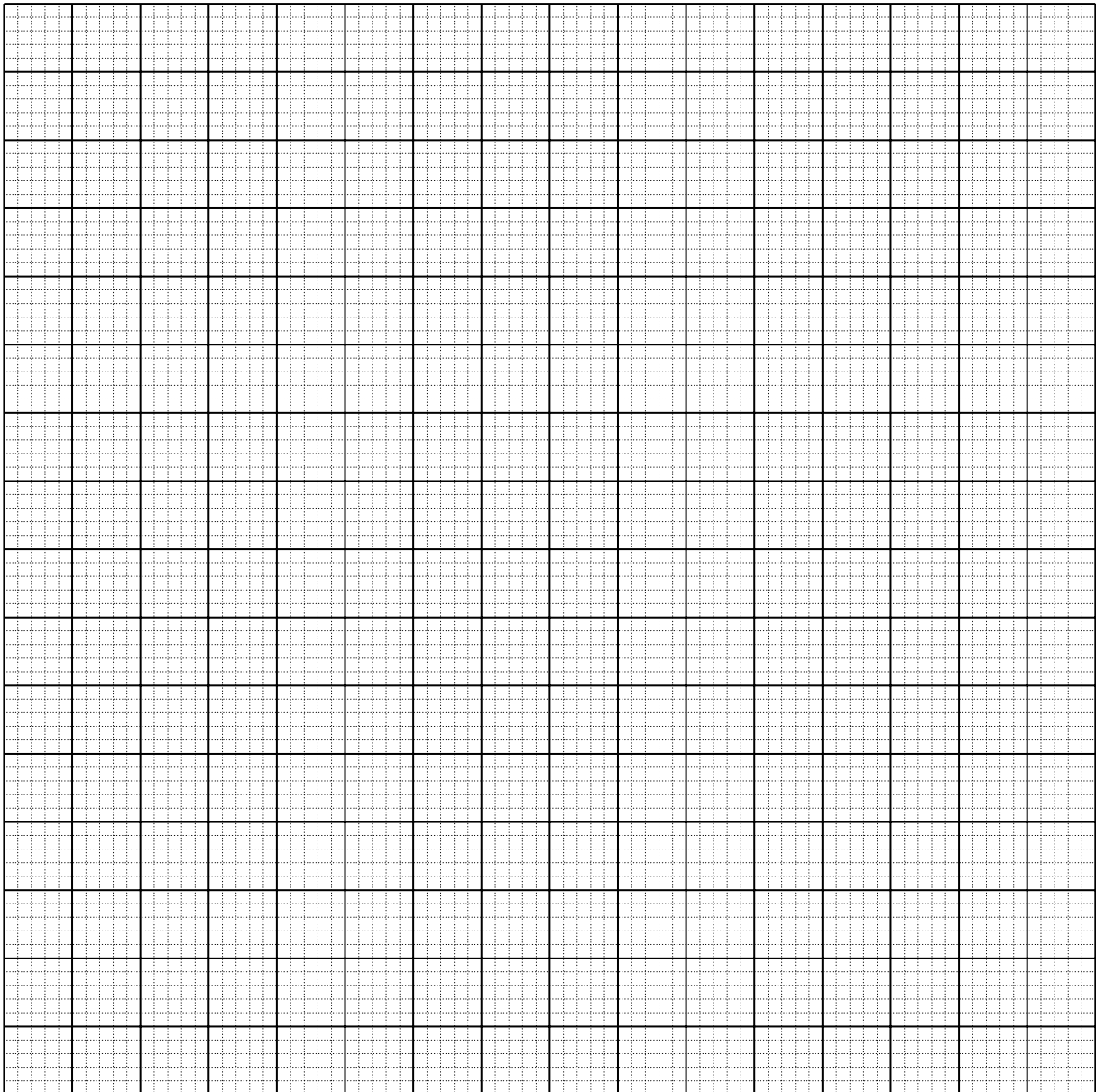
Lipid: _____

Sugar: _____

[2 marks]

Total 16 marks

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END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.